

**SafePoint: An Interactive Incident Map Reporting System for Assisting
and Raising Awareness of Potential Hazards among Civilians**

**A Thesis Project
Presented to the Faculty of the
Information and Communications Technology Program
STI College Novaliches**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

**Justin Allen T. Azucena
Jan Lloyd Y. Babiera
Jareen P. Bagay
John Kenneth H. Fetalvero
Fransee T. Rodriguez**

November 28, 2025

INTRODUCTION

Project Context

In recent years, technology has advanced in parallel to the infrastructure of the modern era. A time where technology evolves every year and buildings get complex and intricate, we are now in a fast-evolving time. With the rise of modern infrastructure comes the cost of maintaining safety and risk-free citizens when it comes to public safety. According to Vergel-Tovar, et al. there is a relation between the rise of traffic incidents in areas with a dense and urban design. This study was conducted in Bogota, Colombia where it was built as a car-centric before the implementation of the Bus Rapid Transit System. For cities who have a high traffic of vehicles, it is a challenge to maintain safety and prevention of hazards as society continuously evolves every moment. This study not only applies to Columbia, but to other car-centric countries or cities where public safety is a priority as high density vehicles can tend to cause different hazards. The rise of modern infrastructures attracts a lot of public traffic, from people travelling to go to work, or to people passing by the nearby city. Roads are becoming the main route of transport in modern society, as a majority of citizens own or travel using a motorized vehicle. It is important to maintain the condition of the roads that are being used everyday, as with the amount of vehicles that pass through main roads in cities worldwide, the condition of the road quickly deteriorates especially with heavy vehicles passing through. It raises the concern on how the community can uphold the safety of these roads as the environments change, as the vehicles change.

The proponents noticed the same pattern of concerns in their target location which is Quezon City. With a high traffic of people walking on the side of the streets, people commuting on public transportation, people travelling on private vehicles. This raises a high concern on upholding public safety since the roads in Quezon City are not at their best conditions, with potholes and road cracks visible any time of the day. It poses a hazard warning for the safety conditions of the road in this city. These road hazards affect the flow of traffic around Quezon City because vehicles tend to find alternate routes due

to the visible potholes. It affects not only the people driving, but also the people passing by these main roads. This issue is not just a concern here in Quezon City, but in other places too. According to Pena-Caballero et al. accidents caused by road hazards can be prevented by monitoring potential hazards and conducting regular inspections done by road management agencies. With each road in the US increasing each year as the global economy grows, it also increases the chance of potential traffic accidents occurring. This raises awareness to local government units to dispatch agencies to handle the inspection and maintenance of road qualities to lower the chance of accidents caused by road hazards to happen. This project was created to create a solution to this problem, to create a mode of communication to raise awareness for the citizens who are lacking information when it comes to potential hazards that they may encounter in an urban area. There are many urban areas that lack a dedicated platform to report hazards to the community as well as the local government units. This can impose a heightened risk when it comes to road accidents caused by hazards like, potholes, deterioration of roads, etc..

In response to these concerns happening and to benefit the community of Quezon City. SafePoint was developed to provide the citizens of Quezon City with a platform aimed to assist in disseminating information about local road hazards to spread awareness of potential risks. SafePoint focuses on assisting commuters and civilians who don't have access to private vehicles. The website will use crowdsourced data from the first hand witnesses of these road hazards. The platform will offer an interactive map wherein the users are able to pinpoint the location on the map where they witnessed the hazard. SafePoint promotes the importance of addressing these public hazards to improve the safety of the community of Quezon City. The platform will be operated by a moderator who will oversee the reports and takedown any potential fake reports that can confuse the users of the platform. The vision of this project is to expand outside of Quezon City in order to help raise awareness nationwide as a majority of the people in the Philippines are public commuters. Through the use of SafePoint, the project aims to create a safe and knowledgeable community which can help alert local government units on the number of potential hazards within the vicinity.

Purpose and Description

The purpose of this proposed system is to develop a platform designed to improve the process of identifying and addressing road hazards for local citizens by offering an accessible, community-centered system for viewing, reporting, and sharing critical safety information. Specifically, the project seeks to address the lack of a centralized and reliable mechanism for civilians to communicate real-time road issues within their area. This gap often results in unreported hazards, delayed responses, and preventable accidents. By introducing a structured reporting system, the project aims to provide timely and credible information that enhances public awareness and supports safer travel decisions for all users.

This platform enables individuals to create detailed reports about various road hazards they encounter, such as potholes, damaged pavements, malfunctioning traffic signs, or other environmental dangers that may pose risks to drivers and pedestrians. Each report contains essential information, including the type of hazard, its exact location, the time it was observed, and any supporting images, that helps ensure clarity and usefulness. Once submitted, reports undergo a review process managed by an assigned moderator. The moderator evaluates each entry to verify its legitimacy, prevent misinformation, and maintain the overall integrity of the system. Only verified reports are displayed on the platform's interactive map, allowing users to access a trustworthy and continuously updated source of community-shared safety information.

The development of SafePoint will follow the Agile Scrum methodology, which emphasizes iterative progress, continuous feedback, and collaborative teamwork. By breaking the project into manageable sprints and involving stakeholders throughout the process, the team ensures that the system evolves in response to real user needs and feedback. This approach not only enhances the quality and usability of the platform but also supports adaptive problem-solving, timely feature delivery, and effective risk management.

By leveraging Agile Scrum, the system aims to create an intuitive and dependable environment where citizens can contribute to and benefit from collective knowledge about road conditions. The SafePoint system emphasizes the importance of community participation, recognizing that safety is a shared responsibility that relies on timely communication and accessible tools. Through the consolidation of hazard reports and the promotion of cooperative engagement, the system has the potential to significantly reduce accidents caused by unreported or overlooked road issues.

Ultimately, the system seeks to provide a meaningful solution that not only addresses an existing inefficiency in public safety communication but also encourages a culture of awareness, responsibility, and proactive involvement. By improving the way road hazards are identified and shared, the platform aspires to enhance travel safety, support informed decision-making, and strengthen the connection between citizens and their communities.

Objectives

The general objective of the study is to provide a functional Incident Report System for civilians to easily access a platform for reporting hazards within their location. Allows to efficiently raise awareness between people spreading information with each other, this prevents people from finding themselves in an unexpected situation that may harm them. Specifically, the study aims to do the following:

1. To design and develop a user-friendly system that allows civilians to report incidents or hazards within their area.
2. To implement a functional map allowing users to easily navigate through various incidents in and outside their area.
3. To develop a system that can show incidents posted by civilians to be viewed by users, increasing alertness and vigilance.
4. To develop a system that can raise awareness of incoming storms.
5. To develop a system that provides users a platform for interactive discussions about specific incidents.

Scope and Limitations

Scope

The scope of this study explains what the system will do and the specific features it will offer to help people report and stay aware of road hazards in their area.

- The system will serve as a centralized platform for civilians in Quezon City to report road hazards and incidents within their area.
- The system will provide an interactive map that displays verified hazard reports for easy navigation and situational awareness.
- The system allows users to create detailed incident reports, including the hazard type, exact location, date and time observed, description, and supporting images.
- The system will ensure the accuracy and credibility of information through a moderator who reviews and validates all submitted reports before posting them publicly.
- The system allows users to stay informed about potential natural disasters (e.g., storms) by showing related alerts that may impact road safety.
- The system will include a discussion feature where users can share additional insights, updates, or clarifications regarding specific incidents.
- The system allows commuters, pedestrians, and civilians with or without private vehicles to easily access reliable, crowdsourced safety information.

Limitations

This study has some limitations that show what the system cannot do or cover. These help explain the possible problems or limits while developing and testing the system.

- The system will not verify the accuracy of every report automatically.
- It will be limited to one area or community for testing.
- The system will not include emergency response features, only reporting and alerting.

Review of Related Literature/Studies/Systems

Kenzie Kao, Saad Ahmed, Reshma Payla, and Mohammed Alsabri (2022), this study dives into how an incident report system can help prevent major casualties and people dying in incidents for medical reasons. This study points out the importance of an incident report system for hospitals, as it can be used as an emergency platform for people who are currently in or in an impending emergency. They concluded that even though the incident report system has been implemented throughout the years to overcome numerous incident victims across the country, the normalization and standardization of the use of it across the people is necessary as it will move forward in the issue of patient safety. Furthermore, they also stated that a widescale implementation of such a system would help hospitals in helping the people. This study proves that despite continuous effort, a fully utilized and implemented incident report system has not yet been done in multiple countries accessing its full utilization and could be further improved for future progress.

Carlos Pena-Caballero, Dongchul Kim, Adolfo Gonzales, Osvaldo Castellanos, Angel Cantu, and Jungseok Ho (2020), states that maintaining infrastructure quality boosts economic growth and promotes the importance of roads to the productivity of national economies. The research highlights the impact of maintaining road quality to prevent or minimize road accidents caused by potholes, and cracks on the roads. They created an algorithm for detecting and classifying potential road hazards that can assist local authorities by uploading the hazard to their system once detected.

Chup-Yip Hon (2025), this study follows research about occupational safety and also health in laboratories. The study states that laboratories commonly use paper or digital formats for incident reporting, which makes it hard to check follow-up action, group events, and use data efficiently for improving safety. Their model highlights the effectiveness of accessing reports and follow-up incidents to streamline actions in case of an emergency, improving injuries avoidance and preventing casualties. Implementing the model will make it easier for businesses and other areas to keep injuries and dangers to a minimum with little resources.

Perera (2021) developed QuickReporter, a mobile and web-based incident reporting system designed to improve the efficiency of reporting and monitoring public or road-related incidents. The system allows users to easily submit incident details and locations through a mobile application built with the Ionic framework and a web platform developed in PHP. Reported incidents are automatically assigned to relevant personnel and displayed on a map for tracking and management. The study's evaluation showed that QuickReporter enhances the speed, accuracy, and usability of incident reporting compared to traditional manual methods, providing an effective digital solution that strengthens communication between the public and authorities.

Kim, Jeon & Kim (2024) analyze the relationship between crowdsourced user-reported feedback (complaints, hazard reports) and actual traffic accidents. By overlaying geolocated user feedback with historical crash data, and applying spatial autocorrelation and clustering, they demonstrate that areas with high densities of citizen reports also tend to coincide with accident hotspots. Their findings suggest that public feedback can dynamically complement traditional traffic safety data, helping authorities identify emerging risk zones even before formal crash data accumulate. For a citizen-centered road-hazard reporting system, this underscores the potential of leveraging user-generated reports for proactive risk detection.

Balmer, Winters, and colleagues (2023) present *WalkRollMap.org*, a web-based, crowdsourced tool for mapping pedestrian hazards, near misses, falls, and missing amenities. The platform uses a user interface close to Google Maps, is mobile-first, collects demographic data, and lets users submit detailed reports (including location, incident type, and narrative). Visualizations (maps, timelines, word clouds) help communities and local planners understand prevalent hazard types and spot spatial trends. This system is a strong case study for a community hazard-reporting platform, showing how inclusive design and participatory mapping can empower citizens to surface infrastructure issues (dangerous crosswalks, slippery paths) that might otherwise go unreported.

Mangrobang and Genove (2025) describe E-report MO, a mobile application for crime reporting that integrates an interactive map and SMS fallback. The app allows citizens to report incidents in real time, even if they lack reliable data connectivity, via SMS. Law enforcement can then visualize reports on a GIS-based map and respond more efficiently. The authors detail the system's design, user interviews, iterative development, and implementation considerations — underlining the importance of accessible, multi-modal incident reporting tools in public-safety contexts.

Uwho and Asianuba (2025) explore the design and implementation of a real-time, integrated incident reporting system tailored for commuter safety. Their system combines a mobile app and a web interface, which allows commuters to report safety incidents (e.g., harassment, accidents, suspicious behavior) and immediately notify relevant authorities. They highlight the importance of real-time notifications, interactive maps, and a backend that can handle high loads — showing how modern technology can transform commuter safety through crowdsourced incident reporting.

Galila (2023) presents Crimeline, a mobile application built around Geographic Information System (GIS) technology for crime reporting. The app enables citizens to report incidents (e.g., theft, assault) in real time, logs them on a map, and allows law enforcement to visualize crime patterns. By enabling participants to report crime efficiently and authorities to see trends, the system fosters collaborative public safety — giving both communities and police data-driven insights.

Mustafa, Han, Sarkar, and Petrova (2025) develop a secure mobile sensing system that uses smartphone data (sensors) to detect and report *abnormal community incidents* (e.g., accidents, environmental hazards, public gatherings). Their system validates data, classifies incident severity, and forwards relevant alerts to both authorities and citizens. This work shows how a mobile platform can act as a proactive incident-sensor network, empowering citizens while maintaining privacy and trust.

Ochante-Huamaccto, Robles-Delgado & Sierra-Liñán (2022) developed an IoT-based mobile application to improve citizen security by mapping crime and incident alerts. Their system uses mobile devices and a backend to collect, map, and alert on dangerous zones, and in their evaluation they found a 102.7% increase in “reports of dangerous places” and a 23.5% faster response time after deployment.

Ariyaratna (2024) examined a crowdsourced mobile solution for geo-targeted emergency reporting, conducting a usability study and showing that users highly value features such as location-tagged incident reports, multimedia uploads, and community feedback. Their prototype highlights how geo-aware reporting apps can improve situational awareness and community engagement in emergencies.

METHODOLOGY

Technical Background

Technologies to be used

The system was developed using a combination of tools and technologies to ensure a functional and user-friendly web application. For the front-end, HTML and CSS were used to design and structure the user interface, providing a visually appealing and responsive layout. JavaScript and C# were utilized for back-end development, handling the logic, data processing, and interaction between the front-end and the database. The web application was built using ASP.NET Core (Razor), which allowed seamless integration of server-side functionality with the front-end components. Microsoft Visual Studio 2022 served as the integrated development environment (IDE), providing comprehensive tools for coding, debugging, and testing the application. SQL Server Management Studio was employed for database creation, management, and storage of system data.

To facilitate data operations within the system, an API Controller was implemented, enabling secure access, retrieval, and manipulation of database information. Additionally, several external APIs were integrated to enhance the system's functionality: EmailJS and Formspree were used for sending automated emails and generating one-time passwords (OTPs), OpenStreetMap was used to display interactive maps for location-based features, and the Windy API provided real-time weather forecasts. These tools and technologies were selected to ensure the system operates efficiently, is scalable, and delivers a seamless experience for users.

Calendar of Activities

The proponents used the Agile Scrum methodology for the project “SafePoint: Assisting and Raising Awareness of Potential Hazards for Civilians using an Incident Report System”. This methodology is the optimal approach for the development cycle of the project.

The proponents used the Software Development Life Cycle (SDLC) to provide a visualization of the development roadmap of the project from the development of the back-end to the front-end. The web-based platform chosen for the project requires careful planning and execution to fit the dynamic environment of the web.

The Agile methodology is an umbrella term for the continuous development and improvement of the project, by utilizing the SDLC, the project is able to be broken down into different stages of development and create a flexible but detailed plan for the development cycle of SafePoint. The Scrum framework under Agile methodology helps break down the complex project into multiple stages to efficiently allocate the appropriate time and resources into each development stage. The use of sprints allows the project to be assigned to different weekly sprints, with predetermined tasks, each with a common goal each week. The use of SDLC and sprints is to create a guideline in the development of the project with proper organization and coordination between different roles.

Sprint 1	Project Weeks: 1-4
Sprint Goal: To start the development of the system, the overall main layout of the front-end, as well as the basic logical functions of the back-end.	
Product Backlog	Increment
● Implementing the basic functions of the user side of the website. ● Development of the UI of the website. ● Creation of the basic logical functions of the website.	● Start the development of the main system, starting with the user side. ● Basic logical functions ● OTP function for login.

Sprint 2	Project Weeks: 5-8
Sprint Goal: To further develop the core functions of the website, and to integrate different API's for the functionality of the different modules.	
Product Backlog	Increment
● Implementation of the interactive map. ● Implementation of the weather module. ● Aligning the appropriate database to the different modules. ● Creation of the moderator side.	● Development of all the major functionalities of the system. ● Refinement of the logical structures of the database.

Sprint 3	Project Weeks: 9-10
Sprint Goal: To refine the core functionality of the system as well as the missing modules of the system.	
Product Backlog	Increment
• Consistent refining of the core functions. • Alterations to certain aspects of the UI.	• Enhanced visuals of the interactive map. • Functionalities of the comments, and validation module.

Resources

Hardware:

- At least Intel® Core™ i3
- At least 8GB of RAM
- At least 64-bit operating system,x64-based processor

Software:

- Windows 11: The operating system is a stable and compatible environment for software development and deployment.
- Microsoft Visual Studio 2022: Used for coding, debugging, and testing the web application efficiently.
- HTML5: Structures the content and elements of the web pages, including headings, forms, tables, links, and other essential components for web layout and accessibility.
- CSS: Styles the user interface for an appealing and responsive design, controlling colors, fonts, spacing, and layout to enhance usability and visual consistency.
- JavaScript: Adds interactivity/logic and dynamic functionality to the front-end, including form validation, animations, event handling, and real-time updates.

- C#: Handles server-side logic, data processing, and communication with the database, enabling the implementation of business rules, data validation, and secure operations.
- ASP.NET Core (Razor): Integrates front-end and back-end functionality to develop a robust web application, allowing dynamic web pages and seamless data binding between the server and user interface.
- SQL Server Management Studio: Manages database creation, storage, retrieval, and manipulation of system data, supporting structured query operations, relational data integrity, and efficient data management.

Requirements Analysis

1. Stakeholders Identification

- Citizens / End Users: Individuals who use the SafePoint platform to report, view, and share information about road hazards in their community.
- Moderators: Designated personnel responsible for verifying and approving submitted reports before they are displayed on the platform.
- Admins / Project manager: Individuals overseeing the overall functionality and maintenance of the SafePoint platform.

2. Techniques Used for Requirements Gathering and Analysis

- Interviews with Potential Users: Conducted discussions with citizens to understand current challenges in reporting hazards and their expectations from a digital platform.
- Observation and Document Analysis: Reviewed existing hazard reporting processes and tools to identify inefficiencies. Studied municipal safety protocols to ensure alignment with real-world procedures.
- Brainstorming and Focus Groups: Collaborated with moderators and administrators to define functional requirements and verification workflows.
- Scenario-Based Analysis: Developed scenarios illustrating common use cases, such as reporting a pothole or a damaged traffic sign, to test workflow efficiency.

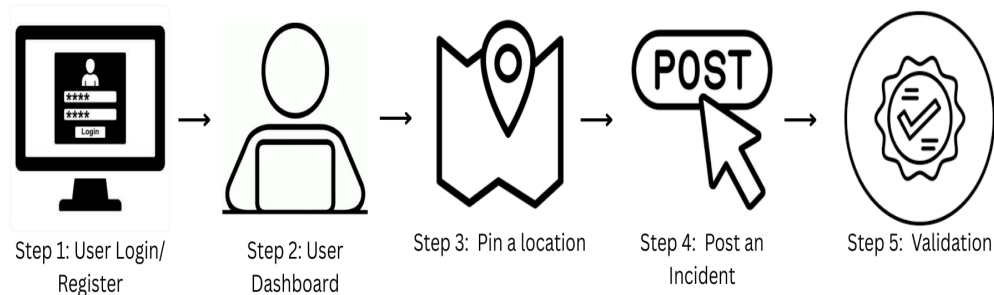
Requirements Documentation

This section defines the system's complete functional scope and non-functional quality constraints, establishing the essential reference for SafePoint's design and implementation. It enumerates all defined features and quality attributes through a functional storyboard and the complete formal specification lists: Functional and Non-Functional Requirements. These components collectively serve as the technical roadmap, explicitly detailing what the system must achieve and how well it must perform, based on the outcomes of the requirements analysis phase.

1. Storyboard

The storyboard visually translates the major functional requirements into a sequential user flow, demonstrating how the system operates from the user's perspective.

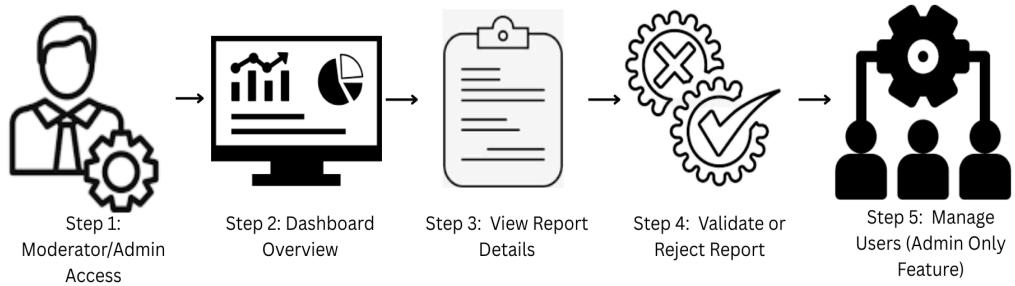
1.1 Incident Report Process



Step	Screen/Module	User Action	System Response
1	Login/Register	User Enters their valid credentials in Login. In Register the user needs to enter one-time password(OTP)	Validates credentials using institution email and password

2	User Dashboard	Users can view all posts and pinned locations that have hazards or incidents in the map. And create posts.	Display all validated incident reports with specific severity. And display all posts and comments.
3	Pin a location	Users will input their locations or pinned it.	The system will automatically copy the address or location and paste it to the posting part.
4	Post an incident	The user will post an incident with specific incident, severity, descriptions, and image(optional).	The system will take the post and send it to the admin/moderator to evaluate the post.
5	Validation	The user will wait for the validation of their posted report.	The system will mark the users post validated or rejected.

1.2 Validation Process (Admin/Moderator)



Step	Screen/ Module	Admin/Moderator Action	System Response
1	Login page	They type their email and password to log in.	The system checks if the login is correct and sends them to the dashboard.
2	Admin/Moderator Dashboard	They look at all reports submitted by users. They can also see which reports are pending, verified, or rejected.	The system shows all incident reports, including the latest ones and flagged reports.
3	Report Information Page	They click one report to view full details like the hazard type, location, description, and picture.	The system displays the complete information of the report.
4	Verification Page	They choose if the report is Valid or Fake. They can also write a short note explaining the decision.	The system updates the report status to Verified or Rejected and sends a notification to the user.

5	User Management Page	The Admin can view users, restrict users who abuse the system, or promote a user to moderator if needed.(Admin Only).	The system saves the changes and updates user access.
---	----------------------	---	---

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

Functional requirements define the specific actions or services that the system must perform. They describe the system's behavior under various conditions.

2.1.1 User Account and Authentication

- User Registration

The system shall allow users to register using an email address and password.

- User Login

The system shall allow users to login using valid credentials.

- OTP Verification

The system shall send a one-time password(OTP) to verify user identity during login or registration.

- Session Management

The system shall maintain user sessions until logout.

2.1.2 Incident Reporting Module

- Create Report

The system shall allow users to submit incident reports, including hazard type, location, date/time, description, and images.

- Map-Based Location Tagging

The system shall allow users to pinpoint hazard locations via an interactive map.

- Upload Images

The system shall allow the upload of one image to support a hazard report.

- Edit Report

The system shall allow users to edit their submitted reports.

- Delete Report

The system shall allow users to manually delete their reports.

2.1.3 Moderator Verification Module

- Report Review

The system shall allow moderators to view, verify, or reject submitted reports.

- report validation Workflow

The system shall update the incident status to verified, rejected, or pending review.

- Remove fake reports

Moderators shall be able to remove false reports or inaccurate reports.

2.1.4 Hazard visualization and Map Module

- Display Verified Hazards

The system shall display only moderator-verified hazard reports on the interactive map.

- Filter Hazard Types

Users shall be able to filter hazards by category.

- Map Interactivity

Users shall be able to zoom, scroll, and navigate through the map.

2.1.5 Weather Tracking

- Weather Integration

The system shall integrate Windy API to provide real-time weather information.

2.1.6 Discussion Module

- Comment on Reports

The system shall allow users to post comments on individual incident reports.

- Discussion Thread Management

The system shall display comments in chronological order.

2.1.7 Admin Controls

- User and Moderator Management

Admins shall be able to create, view, restrict, or remove user and moderator accounts.

- System Monitoring

Admins shall monitor system performance, logs, or flagged incidents.

2.2 Non - Functional Requirements

2.2.1 Usability requirements

- User-Friendly UI

The system must provide a simple and intuitive interface accessible to all civilians.

2.2.2 Maintainability Requirements

- API Maintainability

Integrated APIs must be easily replaceable or upgradeable.

- Modular Code Structure

The system must be developed using a modular architecture for easy updates.

2.2.3 Compatibility Requirements

- Browser Compatibility

The system must run on major browsers (Chrome, Firefox, Edge).

2.3 Operational Requirements

- Internet Connection Requirement

The system requires stable internet connectivity to function..

Design of Software, System, Product, and/or Processes

The design of the SafePoint system focuses on transparency on information dissemination and ensuring that the information present within the system is validated by a moderator or the admin. The system's design is documented through the work breakdown structure, flowchart, entity-relationship diagram to provide a clear visualization of the architecture of the system.

1. System Architecture

The system architecture is built with a focus on:

1.1 Frontend Architecture

The user-side of the system architecture focuses on creating a user-friendly interface while ensuring proper information handling. The system utilizes HTML5 for developing different pages and features of the system styled with CSS, while the JavaScript contains all the logic for each page to call and fetch the API to call the backend controllers and dynamically update the DOM or the Document Object Model.

1.2 Backend Architecture

The server-side, built on ASP.NET Core, serves as the core function of the system, handling all the processing and managing of incident reports. The REST API backend handles incoming HTTP requests and returns structured JSON responses. Data persistence is managed by Entity Framework Core (EF Core), which handles the mapping and interaction with the SQL Server database.

2. Key Architectural Design Components

This section details the critical components implemented across the architecture.

2.1. Authentication and Access Control: The system follows the Role-Based Access Control (RBAC) model to enforce proper authentication and security access to the administrator modules.

2.2 API Controllers: The system uses API Controllers to handle http requests to allow the system to retrieve and send data from and to the database.

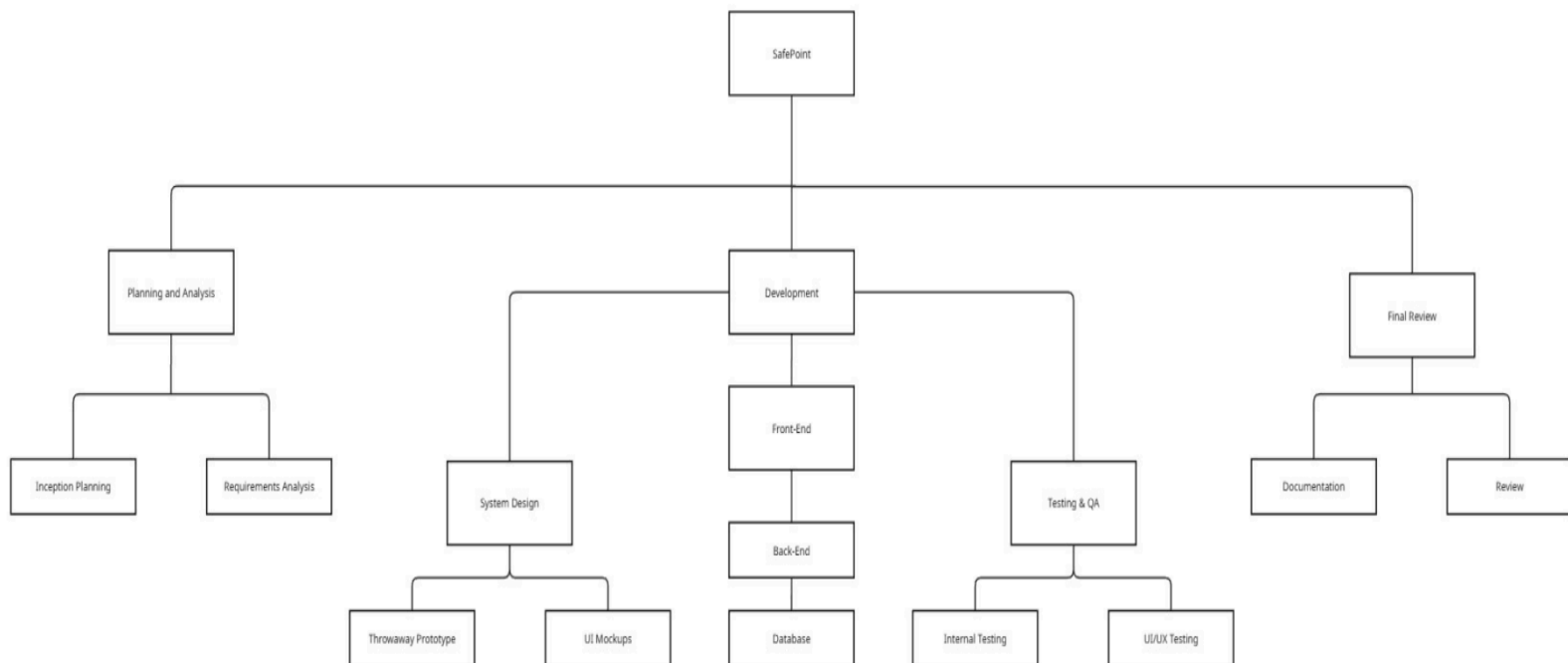
3. User Interface and Experience

The user interface of the system is designed to be user-friendly to all types of users, the UI design is made simple to create a smooth experience. The system UI is centered around the interactive map wherein the users are able to access all the features of the system within the module.

4. Development Approach

The system was developed using an Agile Software Development Life Cycle (SDLC) which allows the development and maintenance of the system during each phase of the development cycle. The development also contains diagrams to visualize the proposed project and guide the development of the system.

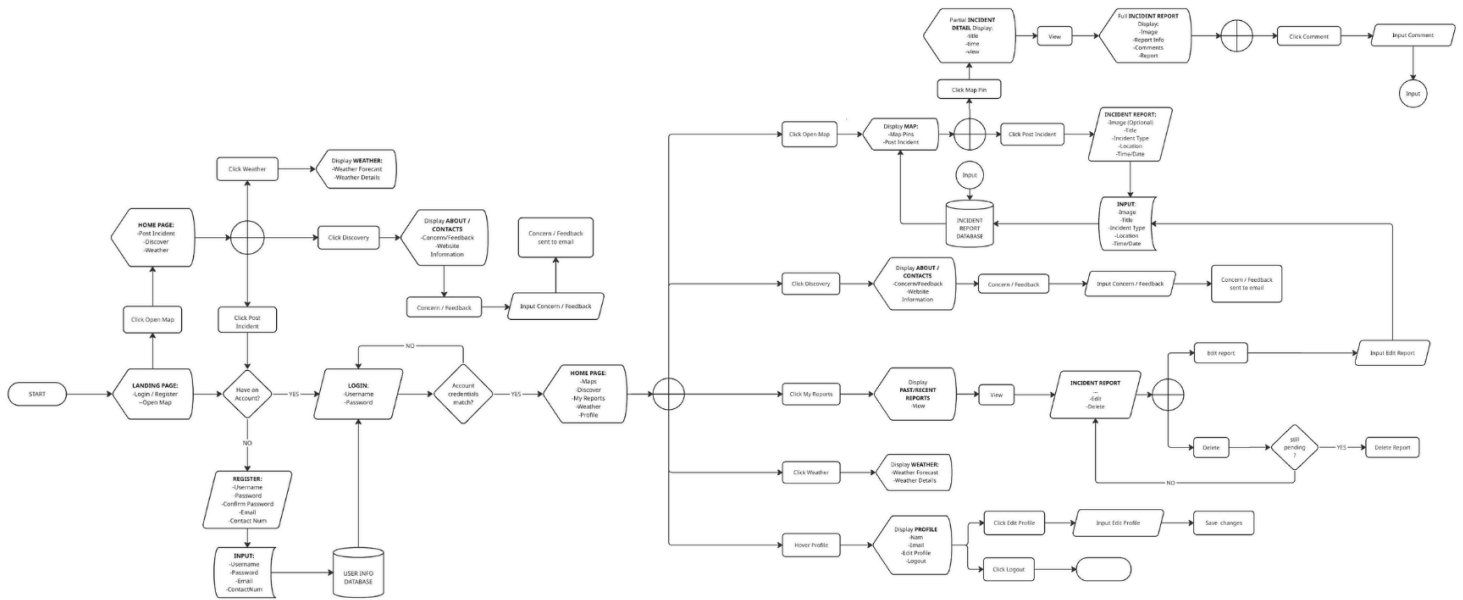
5. Work Breakdown Structure (WBS)



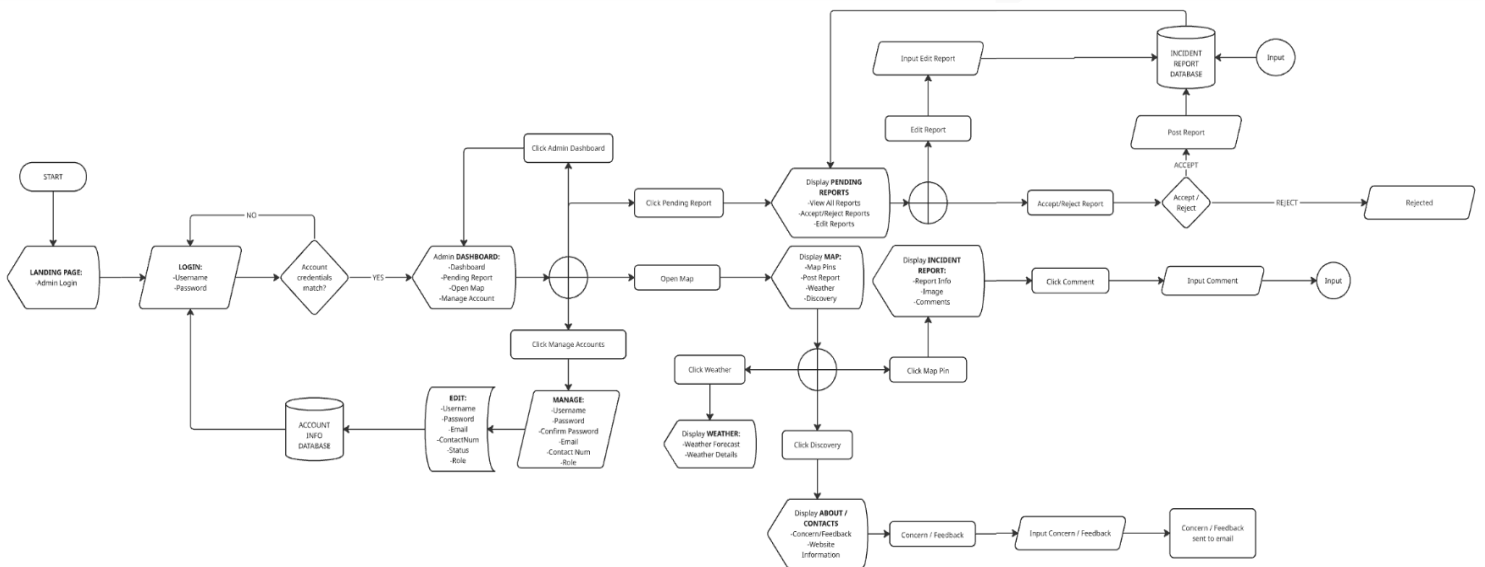
6. Flowchart

6.6 User-side Flowchart

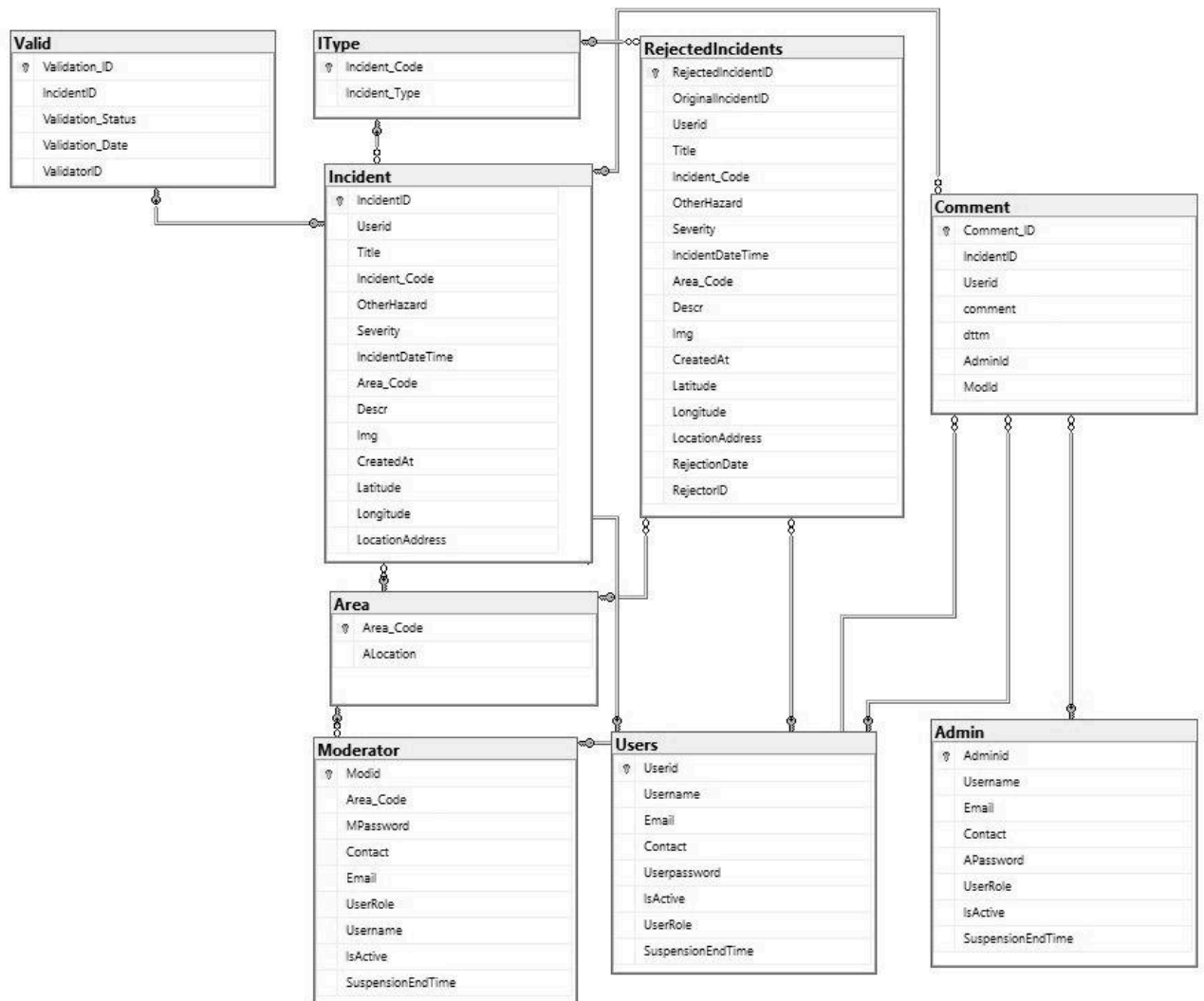
6.6 User-side Flowchart



6.7 Admin-side Flowchart



7. Entity-Relationship Diagram



Development

The development process of the system addresses the importance of creating a user friendly interface that is easy to read and to use. The process includes the following steps:

1. Software Development Life Cycle (SDLC)

The system was developed using the Agile Scrum framework which enables the agile approach to the SDLC. This allowed the SDLC to be split into Sprints segmented by weeks, to be able to create phases in the development cycle, and to be able to emphasize on the different aspects of the system and giving time to resolve any potential issues encountered during the weekly phase.

1.1 Requirements

Requirements of the system were based on research created by other proponents to gather the best possible functionalities.

1.2 Design

The design of the front-end and back-end were designed with the user in mind, this enables any kind of user to be able to use the system effectively. The proponents developed multiple prototypes using Figma before the implementation of the system.

1.3 Implementation

The system used different API's to help develop the core functionalities of the different modules of the website.

1.4 Testing

The website's functionalities and responsiveness is always tested to provide the most efficient service.

1.5 Maintenance

The system will regularly be updated to address any form of bugs or external threats.

REFERENCES

- Kim J. , Jeon W. & Kim S.(2023). Evaluating Road Hazard Maintenance Efficiency Using Citizen Science Data to Improve Road Safety. <https://www.mdpi.com/2071-1050/15/20/15074>.
- Kenzie Kao, Saad Ahmed, Reshma Payla, and Mohammed Alsabri (2022). Incident Reporting Systems: How did we get here and where should we go? A narrative review. https://www.researchgate.net/publication/362590536_Incident_Reporting_Systems_How_did_we_get_here_and_where_should_we_go_A_narrative_review.
- Pena-Caballero C., Kim D., Gonzales A., Castellanos O., Cantu A., & Ho J. (2020). Real-Time Road Hazard Information System <https://www.mdpi.com/2412-3811/5/9/75>.
- Chup-Yip Hon (2025). A Digital Model for Incident Reporting to Support Occupational Safety and Health in Laboratories. <https://www.mdpi.com/2813-8856/2/2/13>.
- Perera, A.S.K. (2021). QuickReporter(Incident Reporting System). <https://dl.ucsc.cmb.ac.lk/jspui/handle/123456789/4406>
- Tovar, Lopez, Lleras, Hidalgo, Rincon, Velasquez & Vega (2020) <https://acrs.org.au/wp-content/uploads/10.33492JRS-D-20-00254.pdf>
- Kim, J., Jeon, W., & Kim, S. (2024). *Analyzing the relationship between user feedback and traffic accidents through crowdsourced data*. *Sustainability*, 16(22), 9867. <https://doi.org/10.3390/su16229867>
- Winters, M., Winters, C., Omstead, K., & McKay, H. (2023). *WalkRollMap.org: Crowdsourcing barriers to mobility*. *Frontiers in Rehabilitation Sciences*. <https://doi.org/10.3389/fresc.2023.1023582>
- Mangrobang, A. N., & Genove, G. P. (2025). *E-report MO: A crime reporting mobile application with interactive map and SMS service*. In *Proceedings of the International Conference on Information Technology and Mechatronics Engineering (ICITME)*, 3287,

Article 030012. AIP Publishing. <https://doi.org/10.1063/5.0262370>

Uwho, S., & Asianuba, I. B. (2025). *Survey of an integrated incident reporting system for commuter safety. International Journal of Innovative Science and Research Technology*, 10(6). <https://doi.org/10.38124/ijisrt/25jun1704>

Galila, G. B. (2023). *Crimeline: Empowering public safety with GIS-based crime incident reporting. International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)*, 3(2).

Mustafa, R., Han, J., Sarkar, N. I., & Petrova, K. (2025). *Secure Cross-Layer Mobile Sensing Framework for Real-Time Disaster Reporting and Visualisation Using a Mobile Application. Sensors*, 25(21), 6766. <https://doi.org/10.3390/s25216766>

Ochante-Huamaccto, Y., Robles-Delgado, F., & Sierra-Liñán, F. (2022). Internet of things based mobile application to improve citizen security. *Indonesian Journal of Electrical Engineering & Computer Science*, 27(1), 386–394. <https://doi.org/10.11591/ijeecs.v27.i1.pp386-394>

Ariyarathna, N. (2024). *Crowdsourced mobile solution to enhance user awareness and response to geo-targeted emergency situations* (Bachelor's thesis). NSBM Green University. Retrieved from ResearchGate. <https://doi.org/10.13140/RG.2.2.22280.43528>